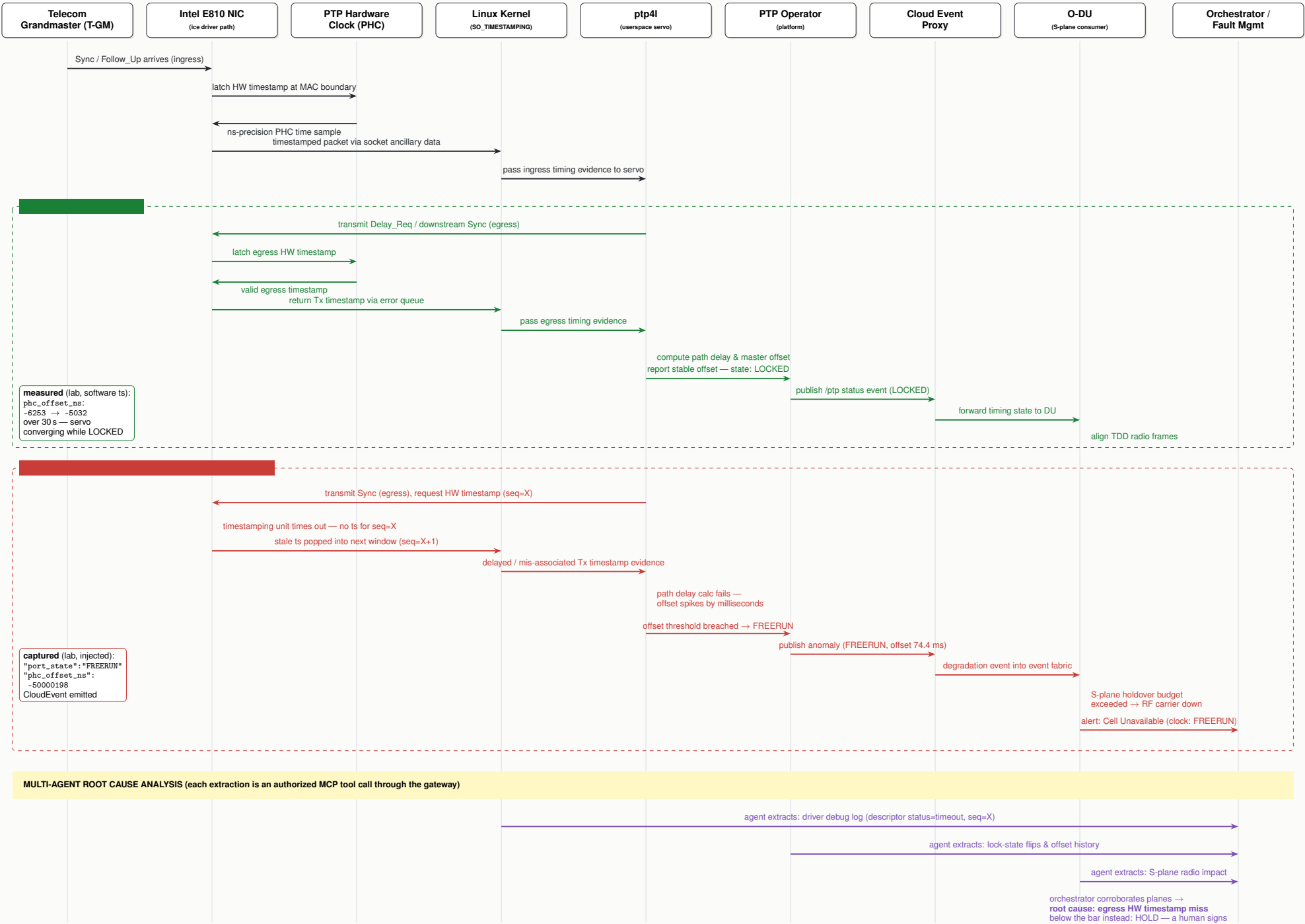


The Timestamp Path

How time crosses hardware, kernel, and userspace — and how one missed egress timestamp takes a cell off the air

Standards: ITU-T G.8275.1 (full on-path support) · IEEE 1588 · O-RAN S-plane · 3GPP TS 28.532 / TS 38.104



Reading this diagram. Time flows down; each lane owns one piece of the truth. The **green** branch is the steady state: every egress packet gets a hardware timestamp, the servo computes a stable offset, and the DU aligns its radio frames. The **red** branch is the same machinery with exactly one thing wrong — the NIC's timestamping unit misses one egress timestamp — and every component downstream behaves *correctly* on the bad input until the cell shuts itself off. The **purple** band is where diagnosis becomes multivendor: the three extractions land in three different organizations' domains, which is why they run as authorized MCP tool calls rather than shared dashboards. The two callout boxes quote this repository's own captures: the servo-convergence series (`data/ptp_offset_samples.jsonl`) and the injected-incident record (`data/incident_capture.jsonl`).